MILITARY SPECIFICATION

AMMONIUM PERCHLORATE, HIGH PURITY

This detail specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the ammonium perchlorate described herein shall consist of this document and the latest issue of MIL-A-82667.

1. SCOPE

1.1 Scope. This specification gives the detailed requirements for two types of ammonium perchlorate with no conditioning agent added.

1.2. Classification. The ammonium perchlorate covered in this specification shall be classified as follows:

Type I Whole crystals (200μm)
Type II Ground crystals (8 to 13μm)

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

RR-S-366	Sieve, Test

MILITARY

MIL-D-3464	Desiccants, Activated, Bagged, Packaging Use and Static Dehumidification
MIL-A-82667	Ammonium Perchlorate, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Division (611), Indian Head, Maryland 20640, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6810

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-1218	ACS Chemicals

PUBLICATIONS

NAVAL SEA SYSTEMS COMMAND (CODE IDENT 10001)

OD 18995	Non-Alkali Metals as Oxides, Determination of
OD 18996	Sulfates as Ammonium Sulfate, Determination of

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Non-Government documents. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM B 330-65(1970)	Average Particle Size of Refractory Metals and Compounds by the Fisher Sub-sieve Sizer
ASTM D 1348-61(1974)	Moisture In Cellulose
ASTM E 203-75	Water Using Karl Fischer Reagent

(Copies of ASTM publications may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19106.)

CODE OF FEDERAL REGULATIONS

49 CFR 100-199	Transportation

(The Interstate Commerce Commission Regulations are now a part of the Code of Federal Regulations (1975 Edition and Revisions) available from the Superintendent of Documents, Government Printing Office, Washington, DC. Orders for the above publication should cite "49 CFR 100-199 (latest revision)").

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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material. The material shall meet the general requirements of MIL-A-82667 and the detailed requirements specified herein. The Type I material shall be solid white crystalline nonspherical granular ammonium perchlorate (NH_4ClO_4) with a nominal particle size of 200µm. The Type II material shall be Type I ground to a particle size range of 8 to 13µm.

3.2 Chemical and physical properties. The chemical and physical properties shall conform to TABLE I.

3.3 Workmanship. The material shall be uniform, free from contamination, foreign material or any other defect that would prevent its use for the purpose intended.

TABLE I Chemical and Physical Requirements

Type I and Type II Composition requirements (wt%)	Min.	Max.	Test par.
Purity	99.3		4.5.1
Chloride, as NH_4Cl	-	0.03	4.5.2
Chlorate, as NH_4ClO_3	-	0.02	4.5.3
Bromate, as NH_4BrO_3	-	0.003	4.5.4
Chromate, as K_2CrO_4	-	0.015	4.5.5
Sulfate, as $(NH_4)_2SO_4$	-	0.20	4.5.6
Sulfated ash, calculated as $NaClO_4$	-	0.25	4.5.7
Iron, as Fe_2O_3	-	0.001	4.5.8
Sodium and potassium	-	0.08	4.5.9
Nonalkali metals as oxide	-	0.04	4.5.10
Moisture, surface	-	0.008	4.5.11
Moisture, total	-	0.05	4.5.12
Water insolubles	-	0.006	4.5.13
Ether soluble	-	0.01	4.5.14
Total volatiles	-	0.04	4.5.15
Stability (hr)	3	-	4.5.16
pH	4.5	5.3	4.5.17
Color	-	-	4.6.1
Friability (% passing no. 100 sieve)			
Type I	-	10.0	4.6.2
Granulation Type I	-	-	4.6.3

TABLE II

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TABLE II Type I Particle-Size Requirements

U. S. Standard Sieve (no.)	Sample passing through sieve (%)	
	Minimum	Maximum
18	100.0	-
20	99.9	-
50	90	96
70	54	70
100	18	30
140	4	15
200	2	5
325	-	2

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. A lot shall consist of the total quantity of cross-blended material. For non-cross-blended material, a lot shall consist of the quantity produced in a single batch, or when manufactured by a continuous process, a lot shall consist of the total quantity offered for acceptance at one time.

4.3 Sampling. A representative sample of each lot shall be selected for testing in accordance with MIL-A-82667.

4.4 Quality conformance tests. These tests shall consist of all the tests described in 4.5 and 4.6 and are subject to the general conditions of MIL-A-82667. When specified in the contract or order (see 6.2), the contractor shall submit a report giving the results obtained for all inspections and a certified statement that the lot meets all the requirements of this specification. Unless otherwise specified all chemicals shall be ACS grade in accordance with MIL-STD-1218.

4.5 Chemical test procedures.

4.5.1 Determination of purity. Ammonium perchlorate purity shall be determined in accordance with the following:

(a) Reagents: Sodium carbonate (Na_2CO_3), anhydrous, analytical reagent grade
Nitric acid (HNO_3), concentrated
Sodium bicarbonate ($NaHCO_3$), reagent grade
Potassium chromate (K_2CrO_4), 5% solution
Standard silver nitrate solution ($AgNO_3$), 0.1N

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(b) Procedure: Weigh accurately about 0.5g of finely ground dry sample into a platinum crucible, and mix carefully with 4.0g of anhydrous sodium carbonate. Overlay the mixture with an additional 1.0g of sodium carbonate. Cover the crucible, and slowly raise the temperature with a Bunsen burner until the contents are completely fused; hold the temperature just above the fusion point for 10 minutes. Allow the crucible to cool. Transfer the fuzed substance quantitatively to a 250-ml wide-mouth Erlenmeyer flask with a minimum quantity of distilled water. Cover with a watch glass, and make the solution slightly acid by slowly adding concentrated nitric acid from a dropping bottle, using litmus paper as an indicator. When all the substances are in solution and the solution is barely acidic, add solid sodium bicarbonate in small portions until the solution is barely alkaline. At this point, the volume should not exceed 75 ml. Add 0.5 ml of potassium chromate indicator, and titrate with standard 0.1N silver nitrate solution. Prepare and titrate a blank containing the quantities of sodium carbonate, nitric acid, and sodium bicarbonate in the same volume of solution as the sample.

(c) Calculation: Calculate the percentage of perchlorate as ammonium perchlorate by the following formula:

$$\text{Percentage of perchlorate as ammonium perchlorate} = \frac{11.75\ (A - B)\ N}{W} - (2.20\ C + 1.16\ D + 0.96\ E)$$

where

A = volume of silver nitrate required to titrate sample, ml
B = volume of silver nitrate required to titrate blank, ml
N = normality of silver nitrate solution
C = percentage of ammonium chloride in sample, as determined in 4.5.2
D = percentage of ammonium chlorate in sample, as determined by 4.5.3
E = percentage of sulfated ash in sample, as determined by 4.5.7
W = weight of sample, g

4.5.2 Determination of chlorides as ammonium chloride (NH_4Cl).

4.5.2.1 The following procedure shall be followed:

(1) Preparation of test sample solution - Dissolve a 1.0-g portion of sample, weighed to 0.1 mg, in approximately 45 ml of water, and filter. Transfer filtrate to a Nessler tube. Add 1 ml of concentrated nitric acid and 1 ml of 2% silver nitrate solution. Dilute to 50 ml and mix thoroughly.

(2) Preparation of comparison standard - Prepare a solution containing 1 ml concentrated nitric acid, 1 ml of 2% silver nitrate solution, and water to dilute to 50 ml.

(3) Procedure - Add standard chloride solution containing an equivalent of 0.000015-g ammonium chloride per ml until the turbidity of the comparison sample matches that of the test sample.

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(4) Calculation - Calculate the percentage of chlorides in the specimen as ammonium chloride by the following formula:

$$\text{Percentage of ammonium chloride} = \frac{100\ (A)\ (B)}{W}$$

where:

A = volume of standard chloride solution required to match sample, ml

B = weight of ammonium chloride per ml of standard solution, g

W = weight of sample, g

4.5.2.2 If the sample contains more than 0.03% chlorides, an additional test shall be performed as follows:

(1) Dissolve a 5.0-g portion of sample, weighed to nearest 0.1 mg, in approximately 45 ml of water, and filter. Dilute filtrate to 50 ml and transfer to a 250-ml glass-stoppered flask. Add 10 ml of 5% (by volume) nitric acid. Pipet 10 ml of 0.02 normal silver nitrate solution into the flask. Add 1 ml of nitrobenzene ($C_6H_5NO_2$) to the flask. Shake vigorously for 1 minute. Add 10 ml of 5% (by weight) ferric nitrate indicator. Back titrate with 0.02 normal ammonium thiocyanate (NH_4CNS) to a pink color which shall persist for no less than 30 seconds after vigorous shaking.

(2) Prepare a blank and back titrate to the same point using the same volumes as in the determination.

(3) Calculate the percentage of chlorides in the specimen as ammonium chloride by the following method:

$$\text{Percentage of ammonium chloride} = \frac{5.35(B - S)\ N}{W}$$

where:

W = weight of sample, g

B = volume of ammonium thiocyanate added to blank solution, ml

S = volume of ammonium thiocyanate added to sample solution, ml

N = normality of ammonium thiocyanate

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4.5.3 Determination of chlorates as ammonium chlorate (NH_4ClO_3) Weigh out 4 or 5 g of sample and dissolve in approximately 100 ml of distilled water in a 600-ml beaker. Add to this solution 50 ml of ferrous sulfate ($FeSO_4$) solution, cover the beaker with a watch glass, and slowly boil the solution for 15 minutes. Remove from heat and place under a lab hood. Carefully and slowly add 10 ml of concentrated nitric acid. Slowly heat solution to boiling point for 5 minutes, cool, and add exactly 20 ml of silver nitrate solution. Add 5 ml of ferric alum indicator solution and 2 ml of nitrobenzene. Titrate with potassium thiocyanate (KCNS) to faint-brown end-point. Prepare a blank and titrate to the same end-point using the same quantities of reagents and the same final volume as in the determination. Calculate the percentage of chlorates in the specimen as ammonium chlorate by the following formula:

$$\text{Percentage of ammonium chlorate} = \frac{10.15N\ (K - k)}{W} - 1.09C$$

where

N = normality of potassium thiocyanate solution

K = volume of potassium thiocyanate solution added to blank solution, ml

k = volume of potassium thiocyanate solution added to sample solution, ml

W = weight of sample, g

C = percentage of ammonium chloride (see 4.5.2)

4.5.4 Determination of bromates as ammonium bromate (NH_4BrO_3). Prepare a sample solution containing 0.100 g/ml by weighing the ammonium perchlorate to the nearest 0.1 mg into a volumetric flask. Dilute to volume. Transfer 100 ml of sample solution to a 500-ml glass-stoppered Erlenmeyer flask, and add 100 ml of freshly-boiled, cooled distilled water. Add 0.5 g of potassium iodide, 5 ml of 1:9 hydrochloric acid (HCl), and 5 ml of 0.2%-starch indicator solution. Mix well and allow to stand in a dark place for 1 hour. Titrate with 0.02 normal standard sodium thiosulfate ($Na_2S_2O_3$) solution until the blue disappears. Run a blank on the reagents. Calculate the percentage of bromates in the specimen as ammonium bromate by the following formula:

$$\text{Percentage of ammonium bromate} = \frac{24.3N\ (V - v)}{S}$$

where

V = sodium thiosulfate required by sample, ml

v = sodium thiosulfate required by blank, ml

N = normality of sodium thiosulfate solution

S = volume of sample solution used, ml

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4.5.5 Determination of chromates as potassium chromate (K_2CrO_4). Dissolve 1 g of sample, weighed to the nearest 0.1 mg, in approximately 95 ml of water, filter, and dilute filtrate in a Nessler tube to 100.0 ml. Add 3.0 ml of a solution containing one part concentrated sulfuric acid (H_2SO_4) to five parts water; mix. Add 1.0 ml of 0.25% diphenyl carbazide solution in alcohol and mix. Prepare a blank in the same manner and add standard (0.0004 g/ml) potassium chromate solution until the color matches that of the sample solution. Calculate the percentage of chromates in the specimen as potassium chromate by the following formula:

$$\text{Percentage of potassium chromate} = \frac{0.04A}{W}$$

where

A = volume of standard potassium chromate solution added to blank, ml
W = weight of sample, g

4.5.6 Determination of sulfates. Sulfates shall be determined in accordance with OD 18996.

4.5.7 Determination of sulfated ash. Sulfated ash, weighed as sodium sulfate (Na_2SO_4) and calculated as sodium perchlorate ($NaClO_4$) shall be determined by the following method. Grind at least 5 g of dried sample to a fine powder. Add 0.2 g of sample to a previously ignited and tared silica crucible. Cover. Heat carefully with a Bunsen flame until decomposition occurs. Continue the ignition of small increments until the entire sample has decomposed and the volatile salts have been driven off. Allow the crucible to cool. Add 3 ml of concentrated sulfuric acid, heat again, and complete the volatilization of sulfuric acid at a dull red heat. Cool in a desiccator and weigh. Calculate the percentage of sulfated ash in the specimen as sodium perchlorate by the following formula:

$$\text{Percentage of sodium perchlorate} = \frac{\text{wt of residue} \times 122.5 \times 100}{\text{wt of sample} \times 71}$$

$$= \frac{\text{wt of residue} \times 172.5}{\text{wt of sample}}$$

4.5.8 Determination of iron as ferric oxide (Fe_2O_3).

(a) Reagents: Hydrochloric acid, concentrated, American Chemical Society (ACS) reagent
Nitric acid, concentrated, ACS reagent
Sulfuric acid, concentrated, ACS reagent
Ammonium thiocyanate, 1% solution of ACS reagent
Iron wire, ACS reagent
Potassium permanganate ($KMnO_4$) 1% solution of ACS reagent

(b) Apparatus: 50-ml Nessler tubes

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(c) Procedure: Weigh accurately 0.10 g of bright iron wire. (An equivalent weight of any other recognized primary standard, with approximately modified preparation procedure, may be substituted for the iron wire standard.) Dissolve the iron wire in 5 ml of concentrated hydrochloric acid. Transfer the solution to a 1-liter volumetric flask and dilute to volume. This is the standard solution. To the ash from the sulfated ash procedure, as specified in 4.5.7, add 1 ml of concentrated hydrochloric acid and 1 ml of concentrated nitric acid. Transfer to a 50-ml beaker with a minimum of distilled water and boil to dryness on a hot plate. Moisten the residue with 2 drops of concentrated hydrochloric acid, and wash with distilled water into a 100-ml glass-stoppered volumetric flask. Accurately transfer a 35-ml portion of the standard solution to a second 100-ml glass-stoppered volumetric flask. Add 2.0 ml of concentrated sulfuric acid, 2 drops of potassium permanganate solution, and 10 ml of 1% ammonium thiocyanate solution to each flask. Dilute to volume, mix well, and transfer to 50-ml Nessler tubes. The sample shall be no darker than the standard.

4.5.9 Determination of sodium and potassium.

4.5.9.1 Preparation of reagents. Quantitatively transfer 1.9066 g of potassium chloride (KCl) and 2.5416 g of sodium chloride (NaCl) into separate 1-liter volumetric flasks containing approximately 250 ml of water. Swirl to dissolve the salt and dilute to volume with water. Each solution contains 1 mg/ml of cation. From these stock solutions, transfer aliquots to other volumetric flasks and dilute with water so that at least four standard solutions are prepared in the range 0.005 to 0.03 mg/ml for the potassium ion and 0.001 to 0.005 mg/ml for the sodium ion.

4.5.9.2 Preparation of standard curves. Determine the net emission in terms of percentage of transmittance of the standard solutions using a wavelength of 766.5 nanometres (nm) for the potassium ion and 589 nm for the sodium ion. The minimum practical slit-width should be used. Prepare a graph either by plotting absorbance versus concentration on linear coordinates or the percentage of transmittance versus concentration on semilog coordinates. Fit a straight line to the points.

4.5.9.3 Analysis of sample. Quantitatively transfer 4.0 g of ammonium perchlorate, weighed to the nearest mg, into a 100-ml volumetric flask. Dissolve in distilled water and dilute to volume. Determine by plotting the percentage of transmittance of each standard solution, a distilled water blank solution, and the sample solution. Read the concentration of the sample directly from the standard curve.

4.5.9.4 Calculation. Calculate the potassium and sodium content as follows:

$$\text{Percentage of potassium and sodium} = 10\ \frac{(R_K + R_{Na})}{W}$$

where

R_K = potassium concentration as read from standard curve, mg/ml
R_{NA} = sodium concentration as read from standard curve, mg/ml
W = weight of sample, g

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4.5.10 Determination of nonalkali metals as oxides. This determination shall be in accordance with OD 18995.

4.5.11 Determination of surface moisture. Surface moisture shall be determined in accordance with ASTM E 203-75 if a direct titration is used and ASTM D 1348-61 if a back titration method is used. In either case a 1:1 methanol:benzene mixture shall be used as the carrier solution.

4.5.12 Determination of total moisture. Total moisture shall be determined in accordance with ASTM E 203-75 using 1:3 methanol:pyridine as the carrier.

4.5.13 Determination of water insolubles. Dissolve 25.0 g of the sample in 175 ml of distilled water and filter the solution through a tared crucible of fine porosity. Wash with distilled water, dry for 1.5 hours at 105°C, cool, and weigh. Calculate the percentage of water insolubles as follows:

$$\text{Percentage of water insolubles} = \frac{(\text{wt of residue, g})100}{\text{wt of sample, g}}$$

4.5.14 Determination of ether solubles as organic material. Transfer 25 ± 0.1 g of the specimen into an extraction thimble of a Soxhlet extraction apparatus. Extract with 100 ml of anhydrous ethyl ether for 2 hours. Transfer ether extract to a tared Vycor dish and evaporate to dryness on water bath. Run a blank determination on 100 ml ether. The increase in weight minus the blank is calculated as organic material as follows:

$$\text{Percentage of ether solubles} = \frac{(\text{increase in wt, g} - \text{wt of blank, g})100}{\text{wt of sample, g}}$$

4.5.15 Determination of total volatiles. Transfer a 10-g portion of sample, weighed to the nearest 0.1 mg, to a tared wide-form moisture dish and weigh. Place in a gravity-convection oven and allow to remain for 2 hours at 177° ± 2°C. Cool in a desiccator and weigh. Calculate the percentage of volatile matter as follows:

$$\text{Percentage of volatile matter} = \frac{(W_1 - W_2)100}{S}$$

where

W_1 = weight of dish plus sample, g
W_2 = weight of dish plus sample after heating, g
S = weight of sample, g

4.5.16 Determination of stability. For heat stability determination, transfer four 10 g portions of sample to 60 mm diameter aluminum weighing dishes which have previously been heated to approximately 700°C and cooled. Place on shelf located 4 inches from top of gravity-convection oven. Maintain oven temperature at 177°C ± 2°C. Remove one sample from the oven after 2 hours and an additional sample every hour thereafter. Cool the sample in a desiccator 20 to 30 minutes. Add approximately 5 g of cooled

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sample to a mixture of 100 ml distilled water, 0.5 g granular potassium iodide (KI), and 5 ml of 0.2%-starch solution. Stir to dissolve ammonium perchlorate. The solution shall remain almost colorless. Immediate appearance of violet or blue indicates decomposition. Repeat the procedure for each sample removed from the oven.

4.5.17 Determination of pH. Add 50 ml of hot, freshly-boiled distilled water to 20.0 g of specimen. Stir and cool to room temperature (20° to 25°C) and allow the precipitate to settle. Measure the pH on a suitable potentiometer having glass calomel electrodes.

4.6 Physical test procedures.

4.6.1 Color. Dissolve a 10 g portion of sample in 100 ml of water and transfer 100 ml of solution to a Nessler tube. Compare with a standard containing 0.005 g of potassium chromate (K_2CrO_4) per liter. The sample shall be no darker than the standard.

4.6.2 Friability. This test shall be performed on Type I material. There is no friability requirement for Type II.

4.6.2.1 Apparatus. The following apparatus shall be required:

(a) Combs gyrating-sifting machine, or equivalent
(b) Two no. 100 U.S. Standard Stainless-Steel Sieves per RR-S-366
(c) One hundred 3-mm glass beads
(d) Stainless-steel cover lid and pan
(e) Two clappers.

4.6.2.2 Procedure. Triplicate tests shall be performed in accordance with the following steps:

(a) Place 100 g of sample on a no. 100 U.S. Standard Stainless-Steel Sieve with stainless-steel pan and cover lid.
(b) Place assembly on sifting machine and shake for 30 ± 1 minutes.
(c) Weigh out 50 ± 0.01 g of sample remaining on the no. 100 sieve and add to the second no. 100 sieve with one hundred 3-mm glass beads. Assemble with cover and pan.
(d) Place assembly on the sifting machine, attach clappers on each side of the sieve, and shake for 30 ± 1 minutes.
(e) Weigh amount of sample passing through the no. 100 sieve.
(f) Calculate as follows and report the average of the tests:

$$\text{Percentage of friability} = \frac{\text{wt passing sieve}}{\text{initial wt}} \times 100$$

4.6.3 Granulation.

4.6.3.1 Type I particle size distribution.

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4.6.3.1.1 Procedure. Particle-size distribution shall be determined in accordance with the following:

(a) Reagent: Tricalcium phosphate (TCP) conditioner
(b) Apparatus:

(1) Sieve shaker: A Tyler "Ro-Tap" or its equivalent with pan and cover and automatic timer
(2) Sieves: One set, including 8-inch diameter, U.S. Standard Sieves no. 18, 20, 50, 70, 100, 140, 200, and 325 conforming to RR-S-366.

(c) Relative humidity: The atmosphere in the room used for the sieving analyses shall be maintained at a relative humidity not greater than 70%. Very moist or very dry atmosphere cause erroneous results, especially with samples containing substantial percentages of fine particles.
(d) Inspection: Sieves shall be inspected daily for visible evidence of damage or failure, especially near the periphery of the cloth, and checked by running a standard sample on which the results shall not vary more than 1.5% from the previously established average value for each sieve. A master set of sieves shall be kept as a standard, to detect wear on the sieves that are in regular use.
(e) Blending: Blend the composite sample carefully in the sample container, and weigh out 100.0 g. Add 1.0 g of TCP conditioner, and mix thoroughly by shaking for 1 minute in a 1-pint glass jar.
(f) Procedure: Assemble the sieves in numerical order from top to bottom, with a pan under the stack. Transfer the material to the top sieve, add the cover to the top of the stack, and place the set in the shaker. Set the timer to run 13 minutes, and start the shaker. When the shaker has stopped, remove the stack of sieves. Starting with the top sieve, brush any material adhering to the bottom of the sieve into the next lower sieve, and transfer the contents to a sheet of glazed paper. Use a brush to remove any material that remains within the sieve. Weigh the retained material to the nearest 0.1 g. Clean the material from the remaining sieves successively in the same manner, adding each fraction to the preceding ones. Record the cumulative weight, which is equal to the percentage of cumulative retention of the original sample (neglecting the conditioner, which is assumed to pass to the pan).

4.6.3.1.2 Calculations.

(a) The calculations for the percentage of sample passing through the sieve shall be as follows:

$$\text{Percentage of weight passing through sieve} = \frac{(\text{wt of sample} - \text{cumulative wt retained}) \times 100}{\text{wt of sample}}$$

(b) The calculations for the percentage of sample retained shall be as follows:

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$$\text{Percentage of weight retained on a given sieve} = \frac{\text{wt on given sieve} \times 100}{\text{wt of sample}}$$

(c) Weighted average shall be calculated as follows:

$$D_w = \frac{(920.5W_{20})+(569.0W_{50})+(253.5W_{70})+(179.5W_{100})+(127.0W_{140})+(89.5W_{200})+(59.0W_{325})+(29.3W_p)}{W_{20}+W_{50}+W_{70}+W_{100}+W_{140}+W_{200}+W_{325}+W_p}$$

where

D_w = weighted average particle size, µm
W_{20} = weight of sample retained on 20-mesh screen, g
W_{50} = weight of sample retained on 50-mesh screen, g
W_{70} = weight of sample retained on 70-mesh screen, g
W_{100} = weight of sample retained on 100-mesh screen, g
W_{140} = weight of sample retained on 140-mesh screen, g
W_{200} = weight of sample retained on 200-mesh screen, g
W_{325} = weight of sample retained on 325-mesh screen, g
W_p = weight of sample on bottom pan, g.

4.6.3.2 Type II particle size. Particle size shall be determined in accordance with ASTM B 330-65.

4.6.4 Visual inspection. All samples shall be examined visually to verify conformance to the workmanship requirement.

4.7 Acceptance criteria. Failure to conform to any of the requirements shall be cause for rejection of the lot.

4.8 Packaging inspection. The packaging, packing and marking shall be inspected to verify conformance with the requirements of section 5. Sampling shall be in accordance with MIL-STD-105 with an AQL of 2.5% defective.

5. PACKAGING

5.1 Packaging and packing. Unless otherwise specified in the contract or order, packaging and packing shall be level C.

5.1.1 Level C. Unless otherwise specified in the contract (see 6.2), packaging and packing of ammonium perchlorate shall be in accordance with standard commercial practice applicable to the type of material. The packaging and packing shall be of such construction and materials that the contents will be adequately protected against loss or contamination. The containers shall conform to 49 CFR 171-179. Dessicant, conforming to Type I of MIL-D-3464, shall be included in containers as specified in the contract (see 6.2).

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5.2 Marking. Marking of containers shall be in accordance with MIL-STD-129 and 49 CFT 171-179 and shall include, but not be limited to, the following:

(a) Title, number, revision letter and date of this specification
(b) Lot number
(c) Contract or purchase order number
(d) Net weight
(e) Material trade name
(f) Manufacturer's name and location

6. NOTES

6.1 Intended use. The material is intended for use as an oxidizer in solid propellant manufacture for the CKU-5/A rocket catapult assembly and other rocket motor applications.

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement requirements:

(a) Title, number, revision letter and date of this specification
(b) Quantity required
(c) Place of inspection
(d) Place of delivery and desiccant required.
(e) Amount of dessicant required (see 5.1)
(f) Packaging requirements if other than as specified (see 5.1)

6.2.2 Contract data requirements. The items of deliverable data required by this specification are cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID *
4.4	Test report	-

* DID's (Data Item Descriptions/DD Form 1664) for the above data requirement are documented in the applicable ADL (Authorized Data List). Such data will be delivered as identified on completed (numbered) DID's when specified on DD Forms 1423 (Contract Data Requirements Lists) and incorporated into applicable contracts.

6.3 Safety precautions. The safety precaution requirements of the "Contractor's Safety Manual for Ammunition, Explosives, and Related Dangerous Material" (DOD 4145.26M) are applicable and should be specified in the contract or order as required by the Armed Services Procurement Regulations (ASPR) 1-323.

NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore" (OP 5) should be made applicable.

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6.4 Suggested source of supply. Products that have met the requirements of this specification in past procurement actions are manufactured by Kerr-McGee Chemical Corporation, Kerr-McGee Building, Oklahoma City, OK 73102. This information is for the convenience of the procuring activity and is not to be construed as a waiver of any requirement of this specification nor as any limitation of additional potential sources of supply.

Custodian:
Navy - OS

Preparing Activity:
Navy - OS

Project Number: 6810-NA40

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-A-82667/1(OS), AMMONIUM PERCHLORATE, HIGH PURITY

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED S/N 0102-014-1802

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